Bantay-Baha

Technical specification

PJDSC 2026

Generated from the pipeline outputs on 24th October 2026

Abstract

Bantay-Baha ranks Philippine provinces for action before a typhoon arrives. It combines the probability R that a province's dengue cases surge in the next four weeks with the share A of its people that the storm cuts off from a hospital, and ranks provinces by the priority $P = R \times A$. This specification describes the data, the method of every layer and how each was evaluated; every result in it is computed from the pipeline outputs at build time. Over 34,050 rolling-origin forecasts in 2018–2019, a multi-quantile gradient-boosted model reaches a relative weighted interval score (WIS) of 0.544 against a seasonal-naive forecast, with a 90% interval coverage of 0.85; its median ensemble with 3 benchmark models reaches 0.444. For the four backtest typhoons, 0.3–1.0 M people lose their route to a hospital on the road and ferry network when roads are cut by flood observed from radar. The lead-lag outbreak graph between provinces keeps 8 links; at one to four weeks its features do not improve the forecast, and the rain and flood features do not either.

Contents

1 Summary **3**
1.1 Findings . . . 3

2 Data **4**
2.1 Acquisition . . . 4
2.2 Harmonised units . . . 4
2.3 Population . . . 4
2.4 Flood-exposure inputs . . . 5

3 Methods **6**
3.1 Layer 1: flood exposure . . . 6
3.2 Layer 2: lead-lag outbreak graph . . . 6
3.3 Forecast and surge probability . . . 6
3.4 Evaluation protocol . . . 7
3.5 Layer 3: access, two methods . . . 7

4 Forecast skill **7**
4.1 What carries the skill . . . 7
4.2 Model comparison . . . 8

5 Surge probability: calibration **9**

6 The lead-lag graph **10**

7 Observed flood: Sentinel-1 **11**
7.1 Checked against Copernicus EMS . . . 12

8 Access loss under four typhoons **13**
8.1 Network against friction raster . . . 13
8.2 Vulnerable groups . . . 15

9 Priority scenarios **15**
9.1 Ompong 2018 . . . 15
9.2 Every storm, 2017–2020 . . . 16

10 Validation, sensitivity and benchmark **18**
10.1 Did high-priority provinces see more cases? . . . 18
10.2 Sensitivity . . . 18
10.3 Metro Manila benchmark . . . 18

11 Leptospirosis: the stretch goal **19**

12 Scorecard against the proposal **20**
12.1 Proposal corrections found in the data . . . 20

13 Limitations and next steps **21**
13.1 Threats to validity . . . 21
13.2 Next steps . . . 21
13.3 Reproducibility . . . 21

14 References **22**

A Acronyms **24**

1 Summary

Bantay-Baha joins three graphs into one ranking: a flood-exposure layer (rain, river routing, observed and mapped flood), a lead-lag outbreak graph learned between provinces, and a road and ferry access graph that a typhoon cuts. A quantile forecast of dengue gives the probability R that a province's next four weeks exceed its usual bad month; two access methods give the share A of people cut off from a hospital; the priority is $P = R \times A$. Dengue forecasting in the Philippines has been studied at province level before (Buczak et al., 2014), and typhoon flooding is followed by leptospirosis (Balbuena et al., 2010); what is new here is the combination with storm-driven access loss, ranked for any typhoon track. Every number in this document was computed by the build (`build.bat`) from the raw files and written into it by `scripts/make_spec.py`.

1.1 Findings

- **The forecast beats the bar.** Multi-quantile XGBoost reaches a relative WIS of 0.42 at one week and 0.66 at four weeks over 34,050 rolling-origin forecasts in 2018–2019 (Section 4).
- **Benchmarked, and beaten on the mean.** Two zero-shot foundation models and a graph neural network reach relative WIS 0.494–0.504; 3 of the 3 have a lower mean WIS than XGBoost. The best mean score is the median ensemble (0.444, 90% coverage 90%); on the median WIS the best is XGBoost + graph (ours) (6.23; the benchmarks 6.54–6.83): the benchmarks win on the high-count epidemic weeks. Storm scenarios use XGBoost, the only model that takes the storm track (Section 4.2).
- **Ablations.** Lead-lag graph features do not help (graph against no graph, relative WIS 0.999), and rain, river, storm and flood features do not help either (1.003). At one to four weeks, recent cases and the season carry the skill.
- **Surge probability R, recalibrated.** Isotonic regression on past rolling-origin forecasts only lowers the Brier score from 0.089 to 0.082 (0.216 for always predicting the base rate), and the largest gap between forecast and observed surge rate in the middle range goes from 21% to 7% (Section 5).
- **The ranking points at the right places.** The Spearman correlation of R with next-four-week incidence is 0.68 for Ompong and 0.73 for Tisoy; 7 of Ompong's ten highest-priority provinces surged against a 40% base rate (Section 10).
- **Access loss, two methods.** For the observed tracks, with roads cut by Sentinel-1 observed flood, 0.3–1.0 M people lose their route on the road and ferry network and 0.6–1.6 M on the friction raster; the gap is reported as a range. Gale swaths cancel 42–100 of 108 ferry routes, and Kendall's τ of the ranking under 9 setting changes is at least 0.99 (Section 8).
- **Leptospirosis, the stretch goal.** In 12 cities, rain is followed by leptospirosis 2–5 weeks later (peak incidence rate ratio 1.40 per +50 mm/week at lag 4); nothing is extrapolated to provinces (Section 11).

> **Not in these results:** System Usability Scale responses from target users (0 so far; the test is built into the dashboard); the official facility registry (hospitals are the OpenStreetMap and hazard-map fallback); see the scorecard against the proposal in Section 12.

Figure 1 shows the forecast target over the whole period, with the four backtest typhoons.

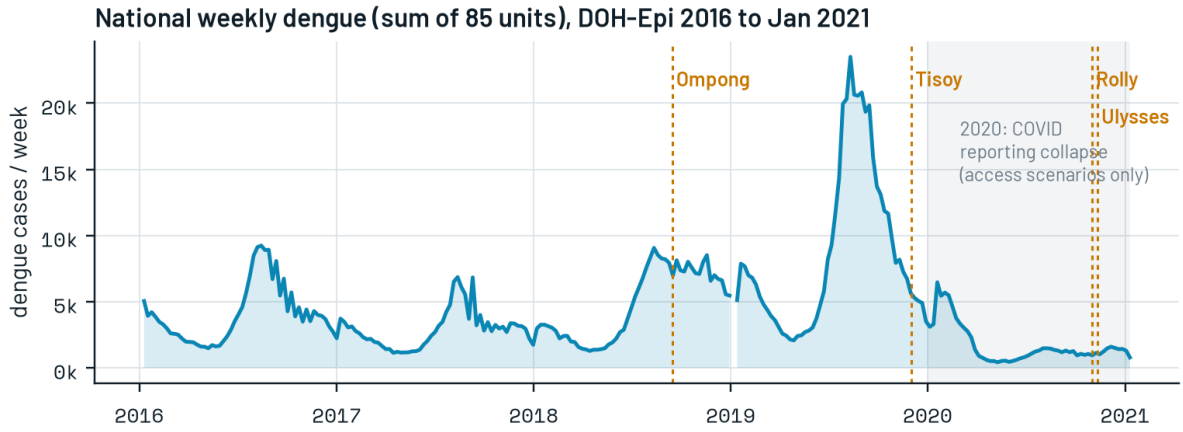


Figure 1: The forecast target: weekly dengue cases from the Department of Health (DOH), summed over the 85 harmonised units, with the four backtest typhoons. 2020 is used for access scenarios only.

2 Data

2.1 Acquisition

501 files from 22 sources (7.0 GB) are listed in `data/raw/MANIFEST.csv` with their size and SHA-256 hash; all were verified on disk before this build. Open sources come from `scripts/download_data.py`, Google Earth Engine sources from `scripts/download_gee.py` (Gorelick et al., 2017), and the team shares them through a private bucket (`setup.bat`, every file checked against the manifest). Table 1 lists them; the two manual sources were not needed (last rows).

The forecast target is the DOH weekly dengue series on the Humanitarian Data Exchange (HDX) (Department of Health Epidemiology Bureau, 2021). Population comes from the 2020 census (Philippine Statistics Authority, 2021) and WorldPop (Tatem, 2017), boundaries and join keys from the common operational administrative boundaries (COD-AB) (Philippine Statistics Authority & National Mapping and Resource Information Authority, 2025), storms from the International Best Track Archive for Climate Stewardship (IBTrACS) (Knapp et al., 2010), rain from CHIRPS (Funk et al., 2015) and IMERG (Huffman et al., 2023), rivers from HydroRIVERS (Lehner & Grill, 2013), roads from OpenStreetMap (OSM) (OpenStreetMap contributors, 2026) and observed flood for Ompong from the Copernicus Emergency Management Service (EMS) (Copernicus Emergency Management Service, 2018).

2.2 Harmonised units

The DOH file has **126 reporting units, not 81 provinces**: provinces mixed with highly urbanised and component cities, plus misspellings (NUEVA VISCAYA, SISQUIJOR, BASILIG CITY for Bislig). COD-AB uses current boundaries (Maguindanao split in 2022, the Special Geographic Area created in 2024). The reviewed crosswalk in `data/crosswalks/` folds cities into their province and undoes the post-2020 changes, giving **85 units**; every dengue unit resolves and every unit has data. A unit-week counts only if every member reported (missing otherwise, never zero). Deaths are not used (the 2016 case fatality of 3.9% is far above the other years), and 2020 fell to 91k cases against 442k in 2019 under COVID-19, so 2020 is never a forecast target.

2.3 Population

The 2020 census table's municipal codes are the old 9-digit Philippine Standard Geographic Code (PSGC) and do not join to COD-AB for the highly urbanised cities, post-split Maguindanao and the Special Geographic Area. The table's own *Province* column already places cities in their province, so units are summed on it: 108.67 M people.

Table 1: Data sources, as recorded in the manifest (files and size on disk).

folder	dataset	role	from	files	MB
dengue_hdx	DOH-Epi dengue, province/city x week 2016–Jan 2021	forecast target	HDX	1	2
census2020	2020 CPH population by municipality	population	HDX	1	1
cod_ab	COD-AB boundaries ADM0–4 (v03, 2025)	join keys, geometry	HDX	2	370
ibtracs	IBTrACS v04r01 Western Pacific	storm exposure, swaths	NOAA NCEI	1	115
chirps	CHIRPS v2.0 daily 0.05°, 2016–2020	rain features, routing	IRI Data Library	5	563
imerg	GPM IMERG V07 daily 0.1°	rain features	Earth Engine	60	68
hydrorivers	HydroRIVERS v1.0 Australasia	river routing	HydroSHEDS	2	139
noah	UP NOAH flood hazard 5/25-yr, 139 provinces	static susceptibility, road proxy	UP NOAH (Drive)	142	4,733
copernicus_ems	EMSR312 / EMSR478 vector packages	observed flood (Ompong AOIs)	Copernicus EMS	16	28
worldpop	WorldPop 2020 100 m UN-adjusted	population grid	WorldPop	1	159
meta_hrsl	Meta HRSL under-5 / 60+	vulnerable groups cut off	HDX	2	112
osm	OpenStreetMap Philippines PBF	road network	Geofabrik	1	611
roroph	roroph RoRo routes (108)	ferry edges (2024–26 proxy)	pinasr/roroph	2	<1
hotosm_health	HOT OSM health facilities	hospital nodes (fallback)	HDX	1	2
noah_facilities	UP NOAH critical facilities	hospital nodes (fallback)	UP NOAH	1	4
map_friction	MAP friction surface 2019	raster access method	Earth Engine	2	2
gfd	Global Flood Database events	context only (ends 2017)	Earth Engine	1	<1
sentinel1	Sentinel-1 scene inventory	pass dates per storm	Earth Engine	1	<1
s1_flood	Sentinel-1 new-flood masks per typhoon	observed flood	Earth Engine	251	2
cchain	Project CCHAIN PIDSR city-week	leptospirosis, 12 cities	HDX	4	7
mendeley_parreno	Parreño national weekly notifiable diseases	leptospirosis, national	Mendeley Data	1	<1
opendengue	OpenDengue V1.3	national cross-check	OpenDengue	3	110
psgc	PSA PSGC 2Q 2026	not needed: COD-AB pcodes resolve every unit	manual	–	–
nhfr	DOH NHFR facility registry	not used: OSM + NOAH fallback	manual	–	–

2.4 Flood-exposure inputs

Figure 2 shows how much of the 85 × 262 panel (22,270 unit-weeks) each input fills. Rain from CHIRPS and IMERG agrees (weekly $r = 0.87$). River routing accumulates CHIRPS rain down the HydroRIVERS network. The flood hazard maps of the Nationwide Operational Assessment of Hazards (NOAH) at a 25-year return period cover 74 of 85 units; the median unit has 20% of its people in a hazard zone. The 11 units with no map are **missing, not zero**: Basilan, Batanes, Biliran, Camiguin, Davao Occidental, Guimaras, Northern Samar, Romblon, Siquijor, Sulu and Tawi-Tawi.

Storm exposure uses every Western Pacific track of 2016–2020 from IBTrACS, with the gale radius from the United States 34-knot radius, else the Japan Meteorological Agency (JMA) 30-knot radius, else a fixed 150 km. The United States radius is missing on 29% of Rolly's gale-strength fixes (JMA fills them); the fixed fallback never applies to the four storms (Section 10).

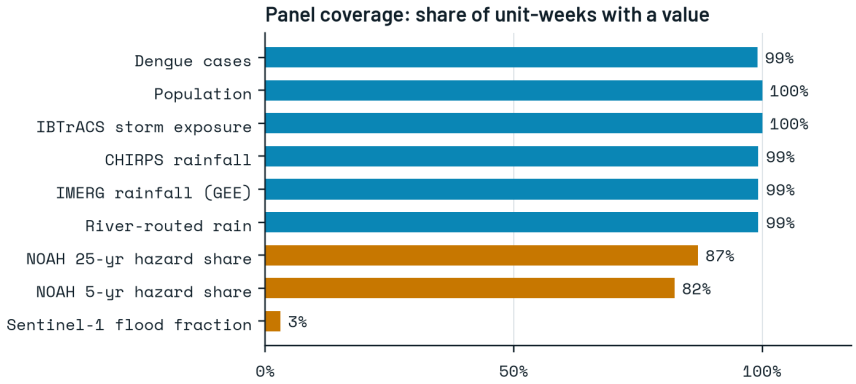


Figure 2: Coverage of the unit-week panel: the share of unit-weeks with a value, per input. Sentinel-1 is filled only for the four landfall windows, by design.

3 Methods

3.1 Layer 1: flood exposure

Each unit-week carries the CHIRPS and IMERG rain of that week and its lags, the upstream-mean rain routed down the river network and weighted by discharge, the storm exposure (whether the unit lies inside a gale swath that week), the share of people in a NOAH hazard zone (25- and 5-year return periods, missing where there is no map) and, for the four backtest storms, the new-flood share seen by Sentinel-1 (Section 7). The Sentinel-1 feature starts in the week that contains landfall plus 10 days, the end of its after-image window, so no forecast origin sees radar from its own future.

3.2 Layer 2: lead-lag outbreak graph

The series is $\log(1 + \text{incidence})$, differenced over 52 weeks to remove the season and standardised inside the training window. PCMCI+, a causal discovery algorithm for time series (Runge, 2020; Runge et al., 2019), runs with partial-correlation tests at lags 1 to 3 weeks and $\alpha_{\text{PC}} = 0.01$; a lagged link between two provinces is kept when its false discovery rate (FDR) q-value is below 0.05 (Benjamini & Hochberg, 1995). On pure-noise checks PCMCI+ without this step returned many false links. PageRank on the reversed graph scores sentinel provinces (Page et al., 1999), Louvain finds communities (Blondel et al., 2008), and lagged cross-correlation with the same FDR filter is the baseline graph.

3.3 Forecast and surge probability

One pooled XGBoost model (Chen & Guestrin, 2016) per horizon (1–4 weeks) is trained with a multi-quantile objective at the 23 quantile levels of the FluSight challenge of the United States Centers for Disease Control and Prevention, on $\log(1 + \text{incidence})$, and back-transformed to cases. The features are lagged incidence, the same week of the previous year, the week of the year, rain and river lags, storm exposure, NOAH hazard, Sentinel-1 flood and leader signals from the graph; Shapley additive explanations (SHAP) give the drivers shown on the dashboard (Lundberg & Lee, 2017). A direct four-week model gives R, the probability of exceeding the unit's 90th percentile of four-week incidence in the training window. R is then recalibrated by isotonic regression (Zadrozny & Elkan, 2002), fitted only on earlier rolling-origin forecasts whose outcome was known at the origin, and only once 850 such outcomes exist.

Benchmarks are scored on exactly the same rows: Chronos-2 (Ansari et al., 2025) and TabPFN-TS (Hoo et al., 2025) zero-shot, each origin's forecast made from that unit's own past only; a message-passing graph network with a long short-term memory (MPNN-LSTM) (Panagopoulos et al., 2021) trained on the same 13-week rolling refits; and the element-wise median of the 23 quantiles of XGBoost and the three (the ensemble).

3.4 Evaluation protocol

Rolling origin, never a random split (Figure 3). From January 2018 to December 2019 the model, the graph and the surge thresholds are refit every 13 weeks (8 refits) on data up to the refit date, and every origin until the next refit is forecast from them. The scores are the WIS on case counts (Bracher et al., 2021), the relative WIS against a seasonal-naive forecast (the same week of the previous year, with its empirical quantiles), the 50% and 90% interval coverage, and an online conformal 90% band built from past errors only (Gibbs & Candès, 2021).

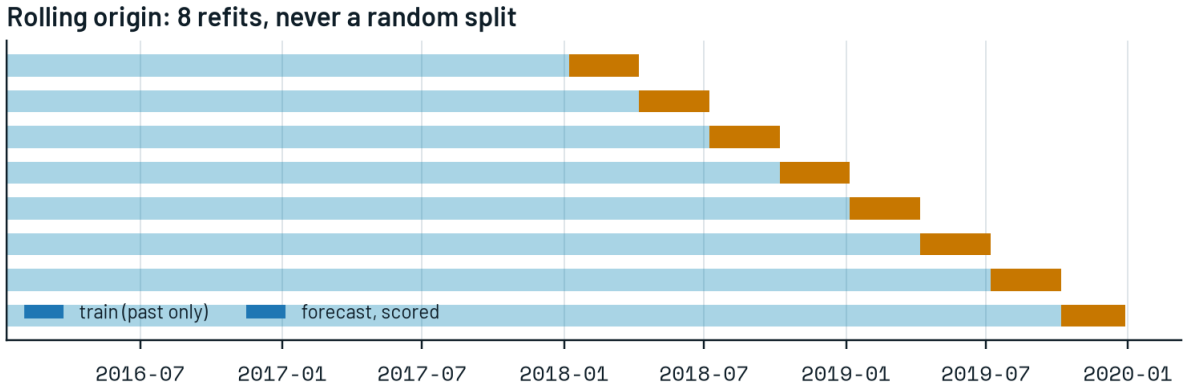


Figure 3: Rolling-origin evaluation: each block is forecast only from a model trained on the data before it.

3.5 Layer 3: access, two methods

Network. OSM highways become a graph of junctions; roll-on/roll-off (RoRo) ferry routes are extra edges (crossing time plus half the headway); hospitals are the Humanitarian OpenStreetMap Team (HOT) health facilities plus NOAH name-matched facilities (2,199). WorldPop is summed to 1 km cells, each joined to the nearest road node, and a multi-source Dijkstra search gives the hours to the nearest hospital.

Raster. A cost-distance search from the same hospitals runs over the motorised friction surface of the Malaria Atlas Project (MAP) (Weiss et al., 2020), with the sea impassable except along the same ferry lines.

Storms apply the same rules to both methods: ferries crossing the gale swath are cancelled; roads (or cells) in observed flood (Sentinel-1, observed tracks) or in NOAH high hazard inside the swath (shifted tracks) are cut, and medium hazard slows them. A cell is cut off if it loses its route, or if its trip grows at least 2-fold and by more than 1 h (after Petricola et al., 2022). The network gives A, P and the action tier; the per-unit minimum and maximum of the two methods give the ranges A_{lo}–A_{hi} and P_{lo}–P_{hi}. Under-5 and 60+ people come from the High Resolution Settlement Layer (HRSL) (Tiecke et al., 2017), summed to the same cells; a cell's vulnerable people are cut off when the cell is.

4 Forecast skill

Table 2 gives the rolling-origin scores of XGBoost by horizon. **The targets are met.** The relative WIS is below 1 at every horizon (0.42 at one week, 0.66 at four) and the 90% coverage is inside the proposal's 80–95% band at every horizon (0.82–0.89; four weeks is close to the floor at 0.82). The 50% intervals cover 0.40–0.47, a little narrow. The online conformal band reaches 0.87, similar to the model's quantiles, a sign that they are not badly miscalibrated.

4.1 What carries the skill

Table 3 compares the full model with its ablations, all scored on the same rows. XGBoost has a 46% lower WIS than the same week of the previous year. The lead-lag graph has 0 links in the 2018 windows and 2–6 in 2019, and its leader features add nothing (relative WIS 0.999). Rain, river, storm and flood

Table 2: Forecast skill of XGBoost by horizon: WIS on case counts for XGBoost, XGBoost without the graph features and the seasonal-naive forecast; relative WIS against the naive forecast and against the no-graph model; empirical coverage.

horizon	*n*	WIS	WIS no graph	WIS naive	rel. WIS	graph / no graph	cov. 50	cov. 90	cov. 90 conformal
1 wk	8,640	14.7	14.7	35.1	**0.42**	0.998	0.47	0.89	0.88
2 wk	8,555	18.1	18.0	35.4	**0.51**	1.002	0.44	0.86	0.87
3 wk	8,470	21.0	21.1	35.6	**0.59**	0.998	0.42	0.84	0.86
4 wk	8,385	23.6	23.6	35.9	**0.66**	0.999	0.40	0.82	0.85
all	**34,050**	**19.3**	**19.3**	**35.5**	**0.54**	**0.999**	**0.43**	**0.85**	**0.87**

features add nothing either (1.003). Two likely reasons, not tested: in a four-year panel rain moves with the season, so the season features already carry it, and any effect at lags beyond four weeks is outside this horizon. Figure 4 shows the forecasts for Cagayan, Ompong's landfall province.

Table 3: Ablations: relative WIS of each comparison on the same rolling-origin rows.

comparison	relative WIS	reading
XGBoost against seasonal naive	0.544	46% lower WIS than the same week last year
with against without graph leader features	0.999	no gain from the graph
with against without rain, river, storm and flood	1.003	0.544 against 0.542 relative to seasonal naive

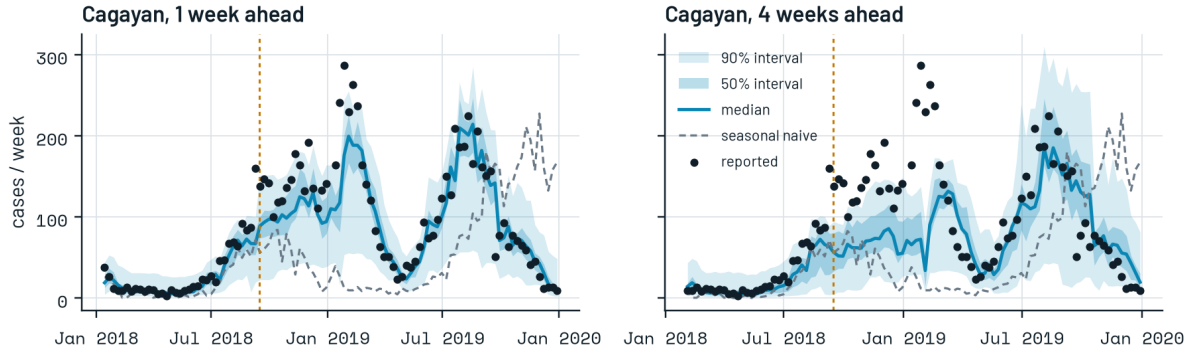


Figure 4: Rolling-origin forecasts for Cagayan, one and four weeks ahead; the dashed line marks Ompong's landfall.

4.2 Model comparison

Every model is scored on the same 34,050 rolling-origin rows (2018–2019, horizons 1–4, WIS on case counts). Table 4 gives the relative WIS against the seasonal naive (below 1 is better) and against XGBoost; Figure 5 shows the same with the coverage by horizon.

- **Mean against median.** 3 of the 3 benchmarks have a lower mean WIS than XGBoost, and the best mean score is the median ensemble. On the median the best is XGBoost + graph (ours) (XGBoost 6.23 cases, the benchmarks 6.54–6.83, the ensemble 6.29). Split at the 90th percentile of weekly cases (198): below it the benchmarks' mean WIS is 1.02–1.08 times XGBoost's, above it 0.80–0.81 times, so the benchmarks' better means come from the large-count epidemic weeks. The ensemble's ratio is 0.94 below and 0.71 above: it gains in both.
- **Chronos-2** as first specified, on log(1 + incidence), collapsed on one-week reporting spikes (relative WIS 112). The headline run uses the incidence scale with past-only covariates; it was

Table 4: Model comparison on the same rows, ordered by relative WIS. Bold: the best model on the mean and the operational model.

model	kind	mean WIS	median WIS	rel. 1 wk	rel. 4 wk	rel. WIS	vs XGB	cov. 50	cov. 90
median ensemble	**ensemble**	**15.7**	**6.29**	**0.331**	**0.555**	**0.444**	**0.816**	**0.48**	**0.90**
Chronos-2, zero-shot	benchmark	17.5	6.54	0.376	0.612	0.494	0.908	0.43	0.88
MPNN-LSTM, graph neural net	benchmark	17.6	6.83	0.384	0.604	0.495	0.910	0.48	0.88
TabPFN-TS, zero-shot	benchmark	17.9	6.72	0.349	0.659	0.504	0.926	0.46	0.87
XGBoost, no rain/river/storm/flood	ablation	19.2	6.26	0.420	0.656	0.542	0.997	0.43	0.85
XGBoost + graph (ours)	**model**	**19.3**	**6.23**	**0.418**	**0.658**	**0.544**	**1.000**	**0.43**	**0.85**
XGBoost, no graph	ablation	19.3	6.24	0.418	0.658	0.544	1.001	0.43	0.85
seasonal naive	baseline	35.5	14.97	1.000	1.000	1.000	1.839	0.42	0.85

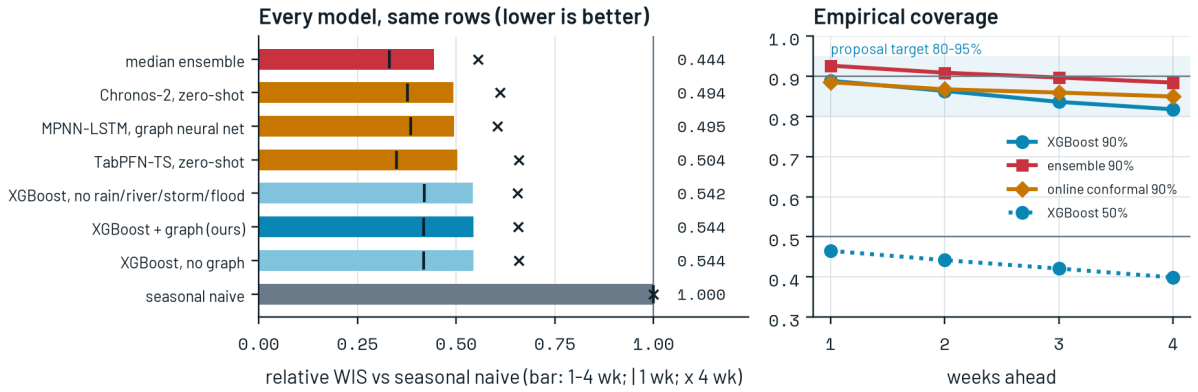


Figure 5: Left: relative WIS of every model over one to four weeks (ticks: one and four weeks). Right: coverage against the nominal lines and the proposal's band.

chosen *after* seeing the first result. Without covariates it scores 0.480, so the covariates do not help.

- **TabPFN-TS** runs univariate (it ignores past-only covariates) with the ungated TabPFN v2 regressor; the default v3.5 weights need a login and licence and were not used. A lagged-covariate variant scores 0.555.
- **MPNN-LSTM** uses border adjacency (133 edges) plus ferry routes (36 more, a 2024–2026 proxy); with no graph it scores 0.492, the same within seed noise, in line with the sparse lead-lag graph.
- **Where it is used.** The benchmarks and the ensemble exist only at the rolling-origin weeks of 2018–2019. Storm scenarios, shifted tracks, 2020 and the storm catalogue use XGBoost with calibrated R, because the zero-shot models take no storm-track input.

5 Surge probability: calibration

The model's raw R is **under-confident in the middle**: in the reliability bins between 0.1 and 0.7, outcomes came true more often than forecast, by up to 21% (Figure 6). Isotonic regression maps raw R to the observed surge frequency. It is fitted at each origin on earlier forecasts whose four-week outcome was already known (target week no later than the origin), so it never sees the future. It needs 850 past outcomes first: the 13 origins before 8 April 2018 keep raw R (1,105 rows). After recalibration the largest middle-bin gap is 7%, smaller than before.

Table 5 gives the Brier scores: recalibration lowers it from 0.089 to 0.082 over all 2018–2019 origins. The base rate is 32%, not 10%, because 2019 was an epidemic year and the thresholds come from earlier,

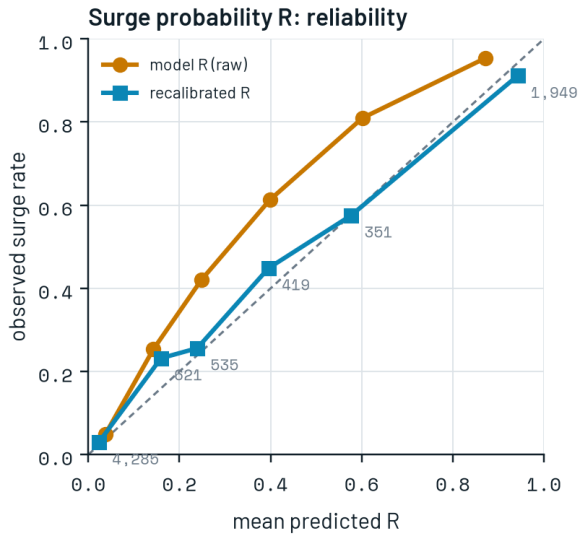


Figure 6: Reliability of R over the 2018–2019 unit-weeks, before (amber) and after (teal) isotonic recalibration; labels give the unit-weeks per bin.

quieter years. The scenario files keep both `R` (calibrated, used for P and the tiers) and `R_raw`. Recalibration changes which provinces reach the provisional tiers (pre-position at $R \geq 0.3$, deploy at $R \geq 0.5$ with $A \geq 10\%$): for Ompong the mean R rises from 0.22 to 0.29. The tier thresholds still need review with users.

Table 5: Brier score of R before and after recalibration, against always predicting the base rate.

rows	n	base rate	mean R raw	mean R cal.	Brier raw	Brier cal.	Brier base rate
all 2018–2019 origins	8,160	0.315	0.254	0.310	0.089	**0.082**	0.216
origins with a calibrator	7,055	0.359	0.286	0.351	0.100	**0.092**	0.230

Ties. Isotonic regression is a step function, so provinces share values: for Ompong the 85 units take 18 distinct values of R and 4 share the top value (0.99). A ranking by R alone has ties; $P = R \times A$ breaks them by access, and the backtest's Spearman correlations use average ranks.

6 The lead-lag graph

In the final window (84 of the 85 units have few enough missing weeks to enter; $T = 156$ weeks after seasonal differencing), PCMCI+ keeps **8** lagged links between provinces at $q < 0.05$ (Figure 7 and Table 6). Plain lagged cross-correlation with the same FDR filter keeps **4,265**, because the 2019 national epidemic makes almost every pair of provinces move together.

The graph's leader features do not lower the forecast WIS (Section 4), so the proposal's sentinel idea is **weakly supported at province-week level**. Likely reasons: weekly province totals average out local waves, four years is short for 84 series, and reporting artefacts dominate small provinces. Synchrony studies have found travelling dengue waves at longer time scales (van Panhuis et al., 2015). The links are statistical dependencies, not evidence of transmission. The MPNN-LSTM benchmark points the same way: its result does not depend on its graph.

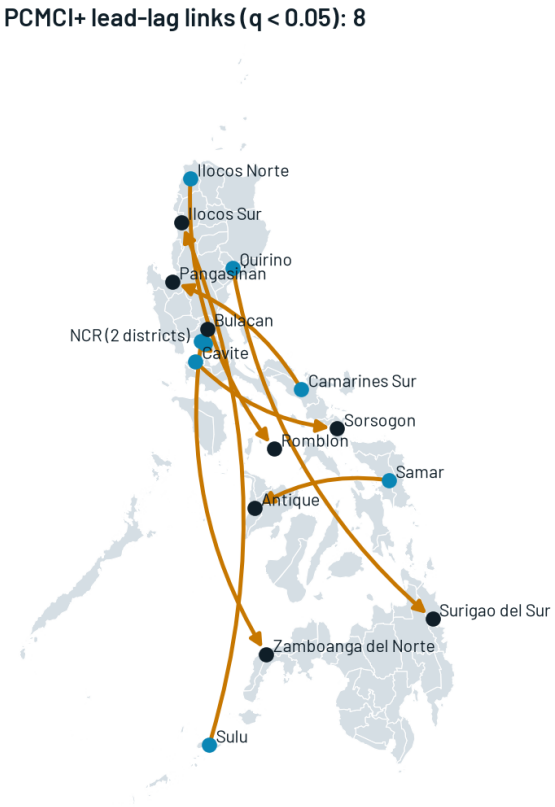


Figure 7: The PCMCI+ links learned on 2016–2019; teal nodes lead.

Table 6: The lagged links PCMCI+ keeps, by strength: lag, partial correlation, FDR q-value and how often the link reappears over bootstrap resamples.

leader	follower	lag (wk)	partial corr.	q	bootstrap freq.
Cavite	Sorsogon	1	0.38	0.001	0.70
Metropolitan Manila Second District	Bulacan	3	0.36	0.006	0.70
Quirino	Surigao del Sur	1	0.36	0.007	0.65
Ilocos Norte	Romblon	1	0.33	0.020	0.75
Metropolitan Manila Third District	Zamboanga del Norte	1	0.33	0.029	0.85
Sulu	Ilocos Sur	1	0.32	0.037	0.55
Camarines Sur	Pangasinan	1	0.31	0.042	0.65
Samar (Western Samar)	Antique	1	0.31	0.045	0.55

7 Observed flood: Sentinel-1

Radar sees through typhoon clouds. For each typhoon, new open water is mapped on Google Earth Engine from the vertical-horizontal (VH) backscatter of Sentinel-1 in 100 m tiles near the track: the before image is the median of days −30 to −3 and the after image the darkest of days 0 to +10, **both from the same relative orbit**. A pixel is new flood when its VH backscatter drops by at least 3 dB and ends at most −18 dB; permanent water, slopes over 5° and specks under 8 pixels are removed (after UN-SPIDER, n.d.). The first version, an after-to-before ratio on decibel values with mixed orbits, recovered almost none of the EMS flood and was replaced.

Figure 8 maps the new-flood share per unit and Table 7 gives the people in new flood per storm. Table 8 repeats this per municipality; there ρ is the Spearman correlation, over the municipalities the radar saw, between network access loss A and the new-flood share. Access loss follows flood but not one-to-one: a municipality can be flooded and keep a dry road to a hospital.

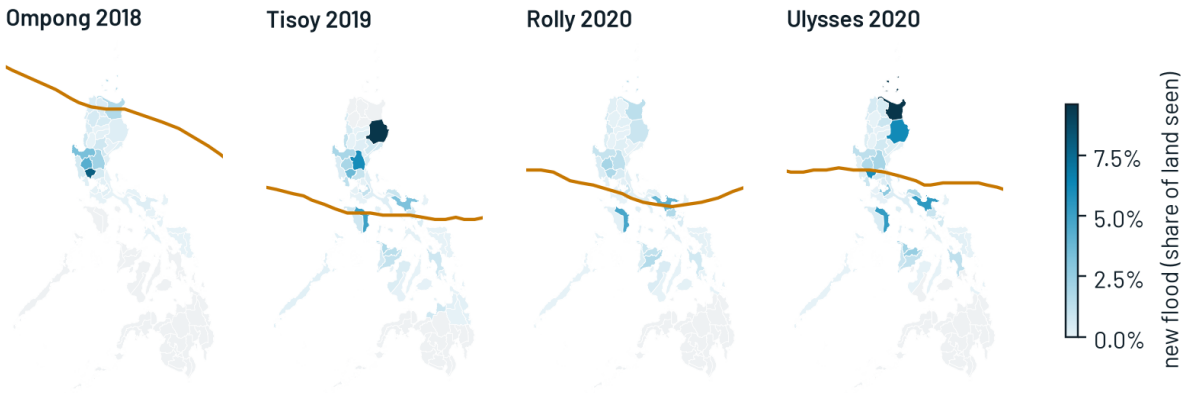


Figure 8: New-flood share of the land Sentinel-1 saw, per unit, with each typhoon's track.

Table 7: People in new flood per typhoon, from the Sentinel-1 masks.

typhoon	people in new flood	units >1% flooded	max unit share	most people flooded
Ompong	0.5 M	5	8.0%	Pampanga
Tisoy	1.1 M	14	9.6%	Isabela
Rolly	0.6 M	10	4.7%	Camarines Sur
Ulysses	1.4 M	14	10.2%	Cagayan

7.1 Checked against Copernicus EMS

Inside the EMS areas of interest for Ompong, EMS mapped 108 km^2 of observed flood and Sentinel-1 flagged 46 km^2; 26 km^2 overlap. **55%** of the Sentinel-1 flood there is confirmed by EMS, and Sentinel-1 recovers **24%** of the EMS area (Figure 9). Low recall is expected: EMS used other imagery and dates, while one Sentinel-1 orbit passed here only twice in the window.

Table 8: Sentinel-1 per municipality: municipalities seen over more than a fifth of their area, municipalities with more than 1% new flood, the three with most people in new flood, and the rank correlation of access loss with flood.

typhoon	seen	flooded	most people in new flood: municipality (province), people, share	$\rho(A, \text{flood})$
Ompong	628	131	Candaba (Pampanga) 42k, 34%; Apalit (Pampanga) 34k, 28%; Arayat (Pampanga) 23k, 18%	0.63
Tisoy	1,074	216	Jaen (Nueva Ecija) 37k, 43%; San Mateo (Isabela) 34k, 47%; Alicia (Isabela) 31k, 62%	0.64
Rolly	1,041	173	Naujan (Oriental Mindoro) 23k, 15%; Nabua (Camarines Sur) 22k, 28%; Bula (Camarines Sur) 21k, 18%	0.54
Ulysses	928	249	Tuguegarao City (Cagayan) 45k, 36%; Solana (Cagayan) 44k, 43%; Apalit (Pampanga) 31k, 28%	0.69

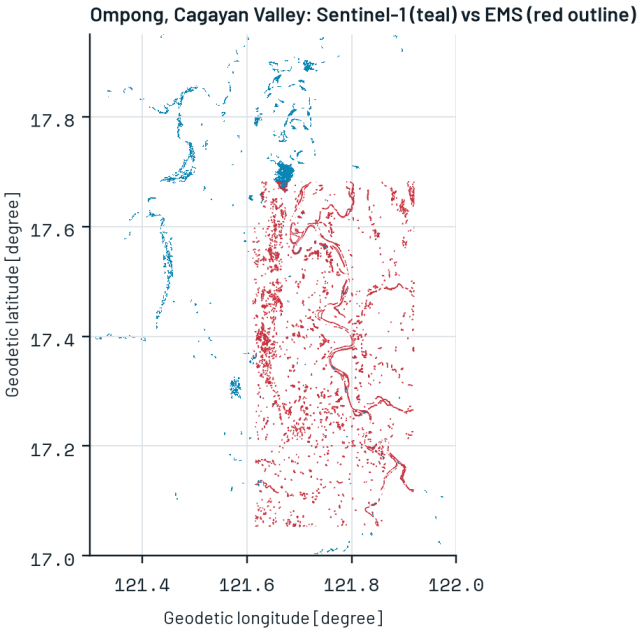


Figure 9: Ompong around Tuguegarao: Sentinel-1 new flood (teal) and Copernicus EMS observed flood (red outlines).

8 Access loss under four typhoons

Baseline. 109.4 M people live in the 1 km cells. On the network, 97.3% of them have a road or ferry route to one of 2,199 hospitals, with a population-weighted mean trip of 0.24 h; the raster reaches 99.5% with a mean of 0.19 h. Results are also written for 1,641 municipalities. Figure 10 shows the ferry routes and hospitals.

Table 9 gives the people cut off by each method for the observed tracks, and Figure 11 maps the network's access loss A per unit.

Table 9: People cut off per observed track: network, raster and the range of the per-unit minimum and maximum of the two; rank agreement of A over units (ρ) and of P (Kendall's τ); top-ten provinces by P shared; vulnerable people cut off (network).

typhoon	network	raster	range	ρ of A	τ of P	top-10 shared	under-5	60+
Ompong	0.3 M	0.6 M	0.3–0.6 M	0.80	0.74	7/10	27k	18k
Tisoy	0.7 M	1.3 M	0.7–1.3 M	0.86	0.72	6/10	65k	41k
Rolly	0.4 M	0.7 M	0.3–0.8 M	0.78	0.70	7/10	34k	22k
Ulysses	1.0 M	1.6 M	1.0–1.7 M	0.85	0.73	7/10	96k	63k

Observed tracks: roads are cut only where Sentinel-1 saw new water. Radar misses floods under dense vegetation, in built-up areas and between passes, so these are **lower bounds**. The 56 shifted tracks have no image and use the NOAH 25-year proxy (every high-hazard road in the swath cut): 2.3–11.3 M people cut off on the network, an **upper bound**. Ferry routes are 2024–2026 operations, a proxy for 2018–2020. A uses radar that exists up to 10 days after the forecast origin: it is a retrospective access estimate.

8.1 Network against friction raster

The proposal runs a raster method next to the network because OSM completeness is uneven, and reports their disagreement as uncertainty. Both use the same hospitals, population cells, ferry lines and storm rules.

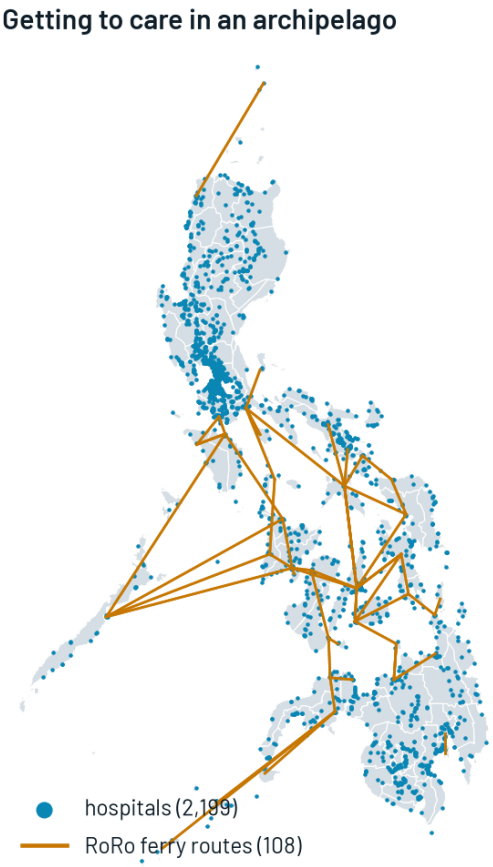


Figure 10: Getting to care in an archipelago: the ferry routes (2024–2026 operations, a proxy for 2018–2020) and the fallback hospital list.

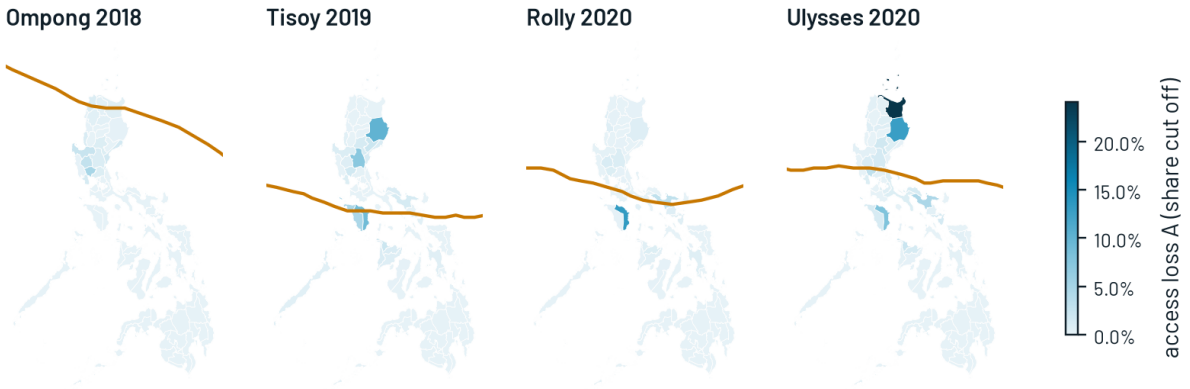


Figure 11: Network access loss A per unit for the observed tracks. Roads are cut by Sentinel-1 observed flood; ferries crossing the gale swath are cancelled.

On a normal day the raster reaches more people (99.5% against 97.3%) and is faster (population-weighted mean 0.19 h against 0.24 h; median cell ratio of raster to network 0.62): it can cross land without a mapped road. The two rank places alike: Spearman 0.89 over the 321,701 cells that both reach, and 0.95 over the 85 units' mean trip.

Under a storm (Figure 12), with Sentinel-1 flood the raster cuts off more people than the network (0.6–1.6 M against 0.3–1.0 M); with the NOAH proxy it cuts off fewer (median ratio 0.66). The gap changes sign with the flood source. A likely reason: at 1 km a flooded cell centre closes the whole cell, while radar flood is patchy and the network closes only the flooded road edges; the proxy closes every high-hazard road, which the raster's cell-centre test samples more sparsely. The unit rankings of A have ρ 0.91–0.99 under the proxy and 0.78–0.86 with radar (they agree better under the proxy), and with radar 6–7 of the top-ten provinces by P are shared. The dashboard shows A and P with this range. On the raster method the sensitivity analysis of Section 10 gives Kendall's $\tau \geq 0.999$.

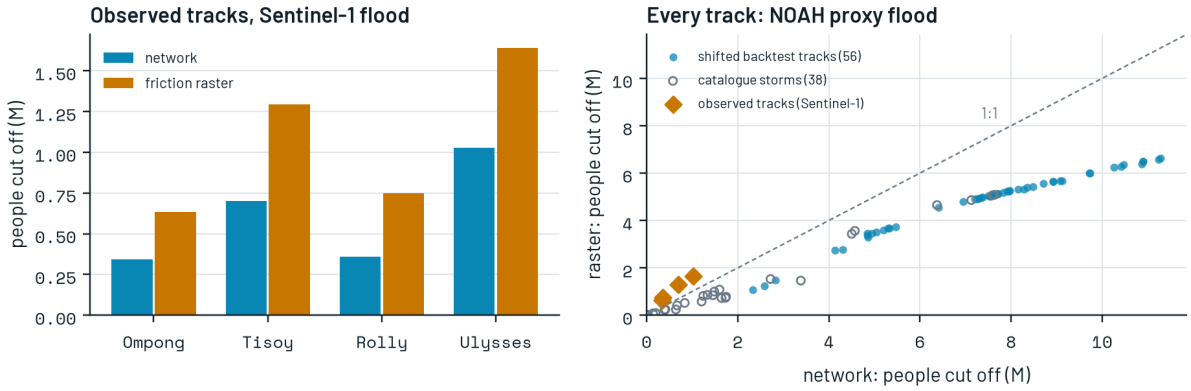


Figure 12: People cut off by each method. Left: observed tracks with Sentinel-1 flood. Right: every track that uses the NOAH proxy (shifted backtest tracks and catalogue storms), with the 1:1 line.

8.2 Vulnerable groups

HRSL puts 11.1% of people under five and 6.7% aged 60 or over; the 2020 census has 10.2% and 8.5%. HRSL therefore **under-counts older people** (and over-counts young children), so the 60+ cut-off numbers are low estimates. Among those cut off, the under-5 share is 7.9–9.5% and the 60+ share 5.2–6.2%.

9 Priority scenarios

Table 10 summarises the four backtest typhoons on their observed tracks: ferry routes cancelled, people cut off on the network, mean A and R over the 85 units, the number of provinces in the two action tiers, and the province ranked first.

Table 10: The four backtest typhoons, observed tracks (local name, with the international name and season).

typhoon	ferry routes cut	people cut off	mean A	mean R	deploy / pre-position	top priority
Ompong (Mangkhut 2018)	42	0.3 M	0.2%	0.29	0 / 22	Pampanga
Tisoy (Kammuri 2019)	100	0.7 M	0.6%	0.12	0 / 12	Camarines Norte
Rolly (Goni 2020)	42	0.4 M	0.3%	0.02	0 / 0	Oriental Mindoro
Ulysses (Vamco 2020)	54	1.0 M	0.8%	0.04	0 / 2	Cagayan

9.1 Ompong 2018

Ompong made landfall in Cagayan on 15 September 2018. The model, trained on data to the landfall week only, puts **Pampanga** first, with 0 provinces in the 'deploy mobile clinics' tier and 22 in 'pre-position

supplies' (Figure 13). 'Deploy' needs at least 10% of people cut off. Of the four observed tracks, with observed flood, 0 put any province there; the hazard-proxy shifted tracks of Ompong put 6–7 provinces there. Figure 14 places every unit on the risk and access axes with the provisional tier thresholds.

The dashboard repeats this for every storm and 14 shifted variants of each track (±0.5 and 1° north or south, ±1° east or west), so a health officer can ask 'what if it comes in further south?' and get an answer in seconds.

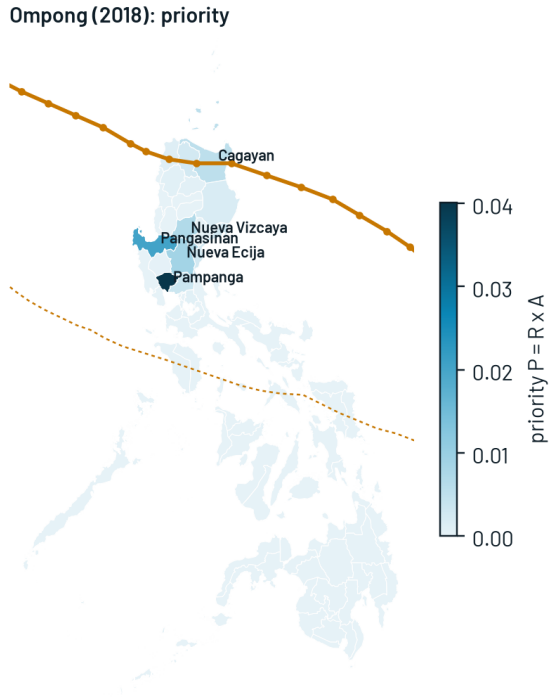


Figure 13: $P = R \times A$ for Ompong with its track and gale swath; the top five are labelled.

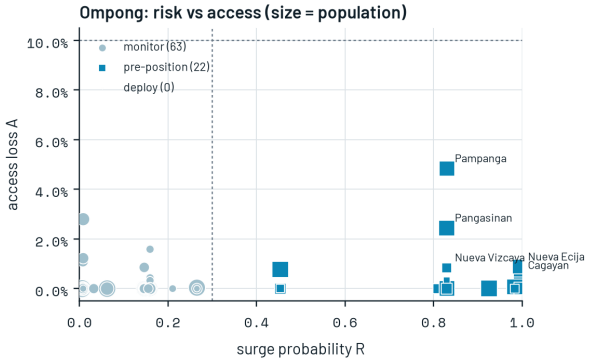


Figure 14: Risk against access for Ompong. Dotted lines are the provisional tier thresholds ($R = 0.3$, $A = 10\%$); marker shape gives the tier.

9.2 Every storm, 2017–2020

Beyond the four backtests, the build scores the observed track of every other storm whose gale swath touched Philippine land: **38 storms**. Each uses the latest model fitted before its week (never later data), R calibrated where enough past outcomes exist (24 of 38), and **the NOAH 25-year proxy for flood**, since no radar map exists for them, so their cut-off counts are upper bounds. 9 storms of 2016 are skipped: their model would have under 52 weeks of history. Figure 15 plots risk against people cut off for all 42 storms and Table 11 lists them.

The largest cut-offs are Kalmaegi 2019, Yutu 2018 and Vongfong 2020; the highest R belongs to Bailu 2019, Podul 2019 and Kajiki 2019, all in the 2019 epidemic season. The network and raster agree across units with a median ρ of 0.96 (minimum 0.82) over the 27 storms with any cut-off.

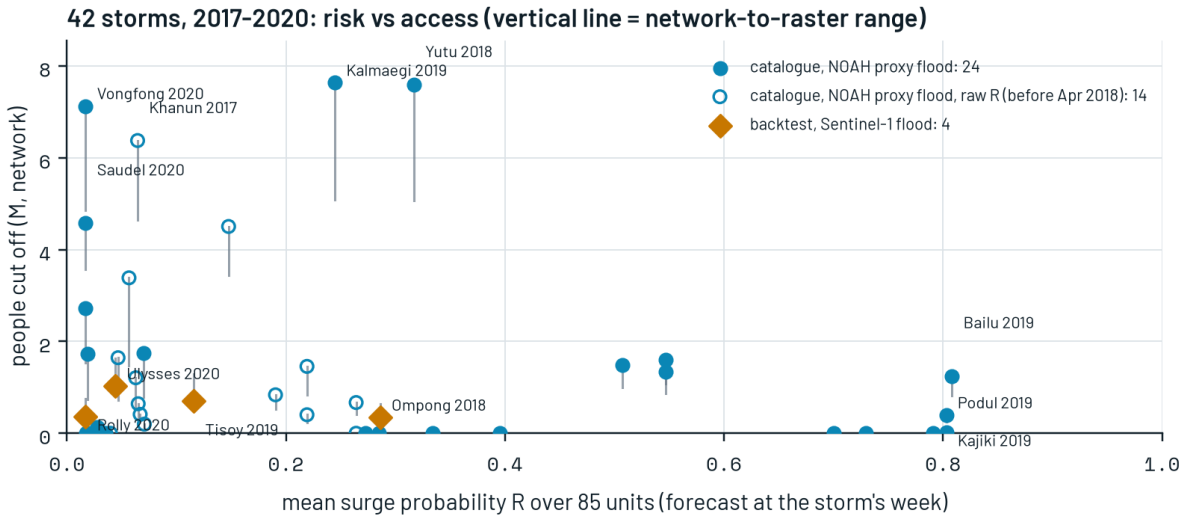


Figure 15: Mean surge probability and people cut off per storm. Diamonds: the four backtests (Sentinel-1 flood, lower bounds); circles: catalogue storms (NOAH proxy, upper bounds); hollow: raw *R*. The vertical line spans the network-to-raster range.

Table 11: Every scored storm in date order. *R*: mean over the 85 units; cut: people cut off (millions), network and raster; *R*×cut: expected people at surge risk who are cut off ($\Sigma\, R\times$ cut off over units); D/P: provinces at deploy / pre-position. ♦ backtest (Sentinel-1); † raw *R* (before calibration was possible).

storm	*R*	cut	raster	*R*×cut	ferry	D/P
Nesat 2017	0.26†	0.00	0.00	0k	2	0/27
Haitang 2017	0.26†	0.67	0.42	174k	2	0/28
Hato 2017	0.22†	0.40	0.23	114k	2	0/21
Pakhar 2017	0.22†	1.46	0.84	419k	2	2/21
Mawar 2017	0.19†	0.84	0.52	368k	2	1/19
Doksuri 2017	0.15†	4.50	3.44	1,883k	22	4/9
Khanun 2017	0.06†	6.37	4.65	799k	24	1/3
Damrey 2017	0.06†	1.21	0.58	24k	64	0/4
Haikui 2017	0.07†	0.20	0.09	6k	0	0/6
Kirogi 2017	0.07†	0.41	0.25	10k	12	0/5
Kai-Tak 2017	0.07†	0.64	0.24	23k	30	0/4
Tembin 2017	0.06†	3.38	1.46	52k	38	0/3
Bolaven 2018	0.04†	0.00	0.00	0k	0	0/0
Sanba 2018	0.05†	1.65	0.73	34k	16	0/2
Gaemi 2018	0.27	0.00	0.00	0k	0	0/29
Son-Tinh 2018	0.55	1.60	1.08	1,452k	2	7/43
Unnamed 2018	0.55	1.34	0.86	1,194k	2	7/43
Barijat 2018	0.40	0.00	0.00	0k	2	0/41
Ompong 2018 ♦	0.29	0.34	0.63	223k	42	0/22
Yutu 2018	0.32	7.59	5.06	2,890k	46	8/18
Usagi 2018	0.33	0.00	0.00	0k	0	0/29

storm	*R*	cut	raster	*R*×cut	ferry	D/P
Danas 2019	0.73	0.01	0.00	10k	2	0/66
Lekima 2019	0.79	0.00	0.00	0k	0	0/72
Bailu 2019	0.81	1.24	0.82	999k	2	6/70
Podul 2019	0.80	0.39	0.24	274k	0	1/73
Kajiki 2019	0.80	0.02	0.01	15k	0	0/74
Tapah 2019	0.70	0.00	0.00	0k	2	0/68
Mitag 2019	0.51	1.49	1.00	266k	2	1/52
Nakri 2019	0.28	0.00	0.00	0k	0	0/31
Kalmaegi 2019	0.24	7.63	5.08	1,957k	46	3/27
Tisoy 2019 ♦	0.12	0.70	1.29	28k	100	0/12
Phanfone 2019	0.07	1.74	0.78	131k	76	1/5
Vongfong 2020	0.02	7.13	4.86	62k	58	0/0
Mekkhala 2020	0.04	0.00	0.00	0k	0	0/2
Bavi 2020	0.03	0.14	0.08	1k	2	0/1
Noul 2020	0.02	0.00	0.02	0k	2	0/0
Linfa 2020	0.02	1.73	0.74	30k	46	0/0
Saudel 2020	0.02	4.57	3.56	49k	2	0/0
Molave 2020	0.02	2.72	1.53	37k	46	0/0
Rolly 2020 ♦	0.02	0.36	0.75	5k	42	0/0
Atsani 2020	0.02	0.00	0.00	0k	2	0/0
Ulysses 2020 ♦	0.04	1.03	1.64	10k	54	0/2

10 Validation, sensitivity and benchmark

10.1 Did high-priority provinces see more cases?

For each typhoon, the ranking made at the landfall week is compared with the incidence each unit saw over the next four weeks, relative to its own threshold (Table 12, Figure 16). The Spearman correlation of R is 0.68 for Ompong and 0.73 for Tisoy: R ranks provinces well for both 2018–2019 storms. P correlates less with cases, as it should: A measures access, not disease. After Ompong, 7 of the ten highest-priority provinces surged against a 40% base rate. After Tisoy 5% of units surged (the 2019 season was ending) and 0 of the top ten, so a low Brier score there says little about the ranking.

Table 12: Backtest per typhoon: units scored, share that surged, Spearman correlation of R and of P with the observed incidence, surge rate among the ten highest-priority units, and the Brier score of R.

typhoon	origin	units	surged	$\rho(R, \text{obs})$	$\rho(P, \text{obs})$	top-10 surged	Brier R	note
Ompong	2018-09-16	85	40%	0.68	0.21	70%	0.161	
Tisoy	2019-12-08	85	5%	0.73	−0.03	0%	0.055	
Rolly	2020-11-01	61	0%	0.41	0.04	0%	0.001	2020: COVID-depressed reporting, indicative only
Ulysses	2020-11-15	85	1%	0.34	0.13	0%	0.021	2020: COVID-depressed reporting, indicative only

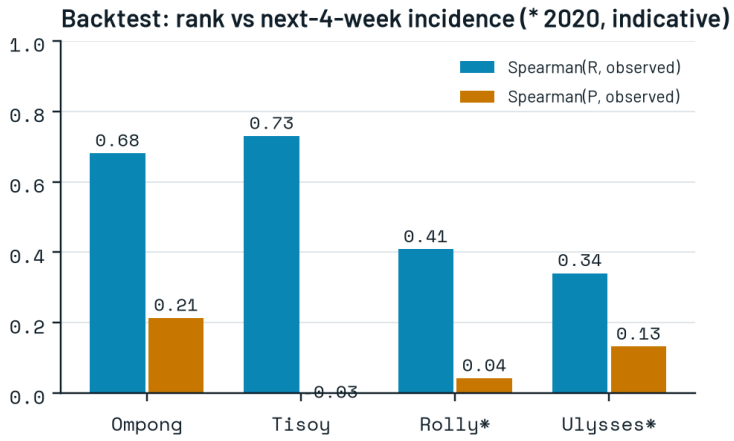


Figure 16: Spearman correlation of each ranking (calibrated R) with the observed incidence over the next four weeks, relative to each unit's threshold. Starred storms (2020) are indicative only.

10.2 Sensitivity

Table 13 gives Kendall's τ between the base ranking by P (network method) and the ranking under each changed setting. The smallest τ is 0.99: the variants change A a little but none reorders the ranking, because the spread of P is dominated by R. The same variants on the raster method give $\tau \geq 0.999$. The gale-radius variants change nothing: the fixed fallback radius never applies to these storms.

10.3 Metro Manila benchmark

Cutting every road of the National Capital Region (NCR) in NOAH 25-year medium or high hazard leaves **29.7%** of NCR without a route within the cut-off rule. A study of the same region reported about 22% of people losing access to higher-level care at a 15 cm flood (Tariverdi et al., 2023). The flood and facility definitions differ, so this is an order-of-magnitude check only.

Table 13: Sensitivity: Kendall's τ of P against the base settings (network method), and Ompong's mean A under each variant.

variant	Ompong τ	Tisoy τ	Rolly τ	Ulysses τ	mean A (Ompong)
cut-off ratio 1.5	1.000	1.000	1.000	1.000	0.23%
cut-off ratio 3	1.000	0.996	0.997	1.000	0.23%
ferry rule: port in swath	1.000	1.000	1.000	1.000	0.23%
gale fallback 100 km	1.000	1.000	1.000	1.000	0.23%
gale fallback 200 km	1.000	1.000	1.000	1.000	0.23%
medium hazard slow x0.25	1.000	1.000	1.000	1.000	0.23%
medium hazard slow x0.75	1.000	1.000	1.000	1.000	0.23%
speed x0.8	1.000	1.000	0.993	0.998	0.23%
speed x1.2	1.000	0.995	0.989	0.999	0.23%

11 Leptospirosis: the stretch goal

The proposal assumed that no open subnational leptospirosis series exists. Project CCHAIN on HDX has weekly counts from the Philippine Integrated Disease Surveillance and Response (PIDSR) system for 12 cities, and a national weekly series of notifiable diseases is published on Mendeley Data. Both are analysed as **retrospective associations**, not forecasts: negative-binomial models of weekly cases on CHIRPS rain 0 to 6 weeks earlier, with seasonal terms, year and city effects and a COVID-19 term. Figure 17 shows the incidence rate ratios (IRR) by lag.

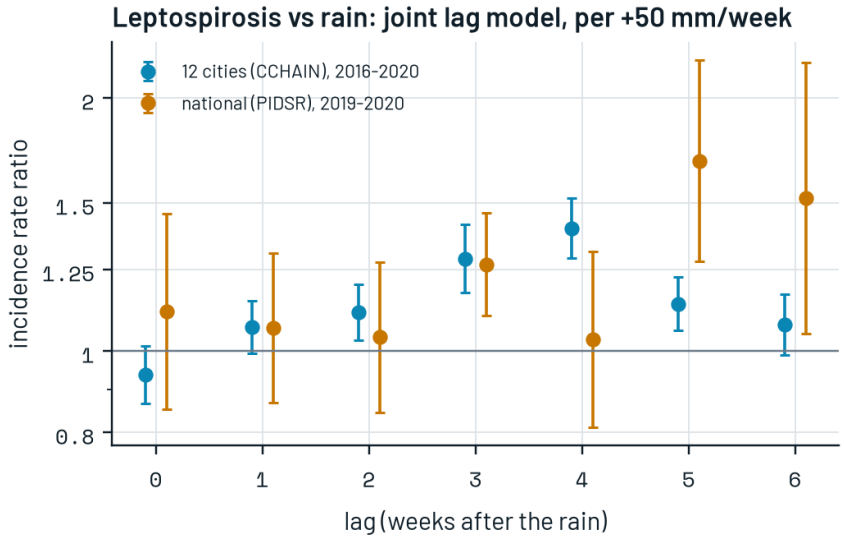


Figure 17: Joint distributed-lag incidence rate ratios (95% confidence intervals) per +50 mm/week of rain, for the cities and the national series.

- **Cities (3,036 city-weeks, 1,685 cases, 2016–2020): the headline.** Rain is followed by leptospirosis 2–5 weeks later (the lags whose 95% confidence interval lies above 1); the joint peak is at lag 4, IRR 1.40 (1.29–1.52) per +50 mm/week, and the cumulative IRR over lags 0 to 6 is 2.45 (2.05–2.92). A heavy-rain week (at or above the city's 90th percentile) peaks at lag 4: 2.00 (1.61–2.48). An earlier Philippine study found a peak after about two weeks (Matsushita et al., 2018); the peak here is later, and the CCHAIN week label is uncertain by about a week.
- **National (80 weeks, 2019–2020).** The joint peak is at lag 5, 1.68 (1.28–2.22); the cumulative IRR is 4.12 (2.41–7.03). With 2019 alone the cumulative effect is 1.51 (0.76–3.00): much of the national signal comes from 2020 (the December surge after Rolly and Ulysses, under COVID-19 reporting).

- **Typhoons.** After Rolly and Ulysses (2020), cities inside the gale swath had 4.31 (1.72–10.85) times the outside cities' ratio of observed to expected cases in weeks 1 to 4, with a pre-landfall placebo of 1.20 (0.32–4.53) (the placebo passes). For Ompong and Tisoy the ratio is 0.81 (0.55–1.20) with a placebo of 2.36 (1.55–3.61): the placebo fails, as weeks of monsoon rain had already raised cases before Ompong. Four storms and 12 cities support no general storm effect.
- **Flood.** Of 48 city-storm pairs, 31 had radar over more than a fifth of the city and 8 show any new flood (at most 2.2%): too few to relate flood to cases, so flood mediation stays an assumption from the literature.

The 12 cities are not a sample of provinces, so nothing here is extrapolated to provinces, and leptospirosis is not in the dashboard's ranking.

12 Scorecard against the proposal

Table 14 sets each objective of the proposal against the result of this build. The status words are decided from the outputs present when the specification is generated.

Table 14: Proposal objectives and the result in this build.

objective (proposal)	result in this build	status	note
Province-week panel 2016–2020 on PSGC: dengue, rain, Sentinel-1, hazard, storms	85 harmonised units × 262 weeks; every listed layer plus river routing	**done**	85 harmonised 2016–2020 units, not 81 provinces (126 reporting units); Sentinel-1 only in the four storm windows
Lead-lag graph (PCMCI+), sentinels (PageRank), communities (Louvain); test sentinel features out of sample	8 links at $q < 0.05$; sentinels and communities on the dashboard; ablation run	**done**	tested and null: graph against no graph, relative WIS 0.999
Relative WIS below 1 against seasonal naive, 2018–2019	XGBoost 0.544; median ensemble 0.444	**done**	
90% interval coverage of 80–95%	0.85 overall; 0.82–0.89 by horizon	**done**	four-week horizon close to the floor
Benchmarks: Chronos-2, TabPFN-TS, MPNN-LSTM; median ensemble; conformal check	all three and the ensemble scored on the same rows; online conformal check	**done**	the ensemble is evaluated, but storm scenarios use XGBoost (the zero-shot models take no storm-track input)
Access loss per municipality, friction-surface and network methods, four typhoons	both methods, 1,641 municipalities, four storms × 15 tracks	**done**	method disagreement reported as the A_{lo}–A_{hi} range
Sensitivity bounds (Kendall's τ); Metro Manila benchmark	two methods, $\tau \geq 0.99$; NCR 29.7% against about 22%	**done**	order-of-magnitude check
Dashboard: ranked provinces, calibrated risk, access, sentinels, drivers, tier, any historical or shifted track	60 backtest scenarios and 38 catalogue storms, pre-computed; R recalibrated	**done**	tiers are provisional rules
Backtest: did high-priority provinces see more cases?	$\rho(R)$ 0.34–0.73 over four storms	**done**	2020 storms indicative only (COVID-19 reporting)
System Usability Scale from three target users	test form and task built into the dashboard; 0 responses so far	**not done**	needs sessions with health and disaster-office users
Leptospirosis (stretch): national lag analysis; region-week series if digitised	national lag analysis, an open 12-city series and a typhoon event study	**partly**	association only, not in the forecast; no province-level series

12.1 Proposal corrections found in the data

- 126 dengue reporting units, not 81 provinces (harmonised to 85); deaths are unreliable; 2020 collapsed under COVID-19.

- An open subnational leptospirosis series exists (Project CCHAIN, 12 cities).
- The EMS activation for Rolly has damage grading only, no flood delineation; the Global Flood Database has no Philippine event after 2017 (Tellman et al., 2021).
- The flood hazard maps come from the NOAH archive at 5 and 25 years, not the national lidar portal; 11 units have no 25-year map.
- HydroRIVERS places the Philippines in the Australasia file, not the Asia file.
- The ferry routes describe 2024–2026 operations with partly coarse port coordinates; the official facility registry blocks scripted download (the OSM and NOAH fallback is used).
- HRSL under-counts people aged 60 or over against the 2020 census (6.7% against 8.5%).
- The proposal's final forecast was the median ensemble; storm scenarios use XGBoost, the only storm-conditioned model.

13 Limitations and next steps

13.1 Threats to validity

- **Two evaluation years.** 2018–2019 contains one epidemic; the best model changes between mean and median WIS because of it.
- **Surveillance artefacts.** Reporting delays and under-reporting are in the target; the forecasts predict reported, not true, cases.
- **Observed flood.** Sentinel-1 sees each place every 6 to 12 days and misses flooding under dense vegetation and in built-up areas; the shifted tracks and the catalogue fall back to static hazard maps.
- **Network gaps.** OSM completeness is uneven outside Luzon (the raster method bounds this); the ferries are 2024–2026 routes; the hospitals are an OSM and NOAH fallback without facility level or capacity. HRSL under-counts older people.
- **Provisional rules.** The tier thresholds, road speeds and the cut-off rule are assumptions, partly addressed by the sensitivity analysis.
- **Causality.** The PCMCI+ links and the leptospirosis lags are associations, not transmission routes or causal effects.

13.2 Next steps

1. Run the System Usability Scale test with three users from regional health offices, local government health units and disaster risk reduction offices (Brooke, 1996), and review the tiers with them.
2. Feed a forecast track from the national weather service and its cone of uncertainty in place of the historical shifts.
3. Add a second radar or optical source between Sentinel-1 passes.
4. Replace the hospital fallback with the official facility registry (facility level, so that access can mean a higher-level hospital).
5. Give the benchmarks storm inputs, or blend the ensemble with the storm signal of XGBoost, so that the scenarios can use the ensemble.

13.3 Reproducibility

`setup.bat` (or `download_data.bat`) → `build.bat` → `run_app.bat`. The evaluation report and the slides are regenerated with `python scripts/make_results.py`, and this specification with `python scripts/make_spec.py`. Missing inputs are skipped, listed in `data/processed/status.json` and on the dashboard; re-running after they arrive needs no code change.

14 References

Ansari, A. F., Shchur, O., Küken, J., Auer, A., Han, B., Mercado, P., Rangapuram, S. S., Shen, H., Stella, L., Shen, X., Turkmen, C., Bohlke-Schneider, M., Erickson, N., Kapoor, S., Li, Z., Wang, B., Williams, A. R., Ahmed, A., Weber, S., et al. (2025). *Chronos-2: From univariate to universal forecasting*. arXiv. https://arxiv.org/abs/2510.15821

Balbuena, A., Herbosa, T., & Gundran, C. P. (2010). Cases of leptospirosis after Typhoon Ondoy (international name: Ketsana) seen at the emergency department: The Philippine General Hospital experience. *WIT Transactions on Information and Communication Technologies*, *43*, 605–612. https://doi.org/10.2495/RISK100501

Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, *57*(1), 289–300. https://doi.org/10.1111/j.2517-6161.1995.tb02031.x

Blondel, V. D., Guillaume, J.-L., Lambiotte, R., & Lefebvre, E. (2008). Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, *2008*(10), P10008. https://doi.org/10.1088/1742-5468/2008/10/P10008

Bracher, J., Ray, E. L., Gneiting, T., & Reich, N. G. (2021). Evaluating epidemic forecasts in an interval format. *PLoS Computational Biology*, *17*(2), e1008618. https://doi.org/10.1371/journal.pcbi.1008618

Brooke, J. (1996). SUS: A "quick and dirty" usability scale. In P. W. Jordan, B. Thomas, B. A. Weerdmeester & I. L. McClelland (Eds.), *Usability evaluation in industry* (pp. 189–194). Taylor & Francis.

Buczak, A. L., Baugher, B., Babin, S. M., Ramac-Thomas, L. C., Guven, E., Elbert, Y., Koshute, P. T., Velasco, J. M. S., Roque, V. G., Jr., Tayag, E. A., Yoon, I.-K., & Lewis, S. H. (2014). Prediction of high incidence of dengue in the Philippines. *PLoS Neglected Tropical Diseases*, *8*(4), e2771. https://doi.org/10.1371/journal.pntd.0002771

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794. https://doi.org/10.1145/2939672.2939785

Copernicus Emergency Management Service. (2018). *EMSR312: Typhoon Mangkhut in the Philippines* [Rapid mapping activation]. European Commission. https://mapping.emergency.copernicus.eu/activations/EMSR312/

Department of Health Epidemiology Bureau. (2021). *DOH-Epi dengue data 2016–2021*. Humanitarian Data Exchange. https://data.humdata.org/dataset/ac63a95e-7296-42fb-802b-7f7541c73e45

Funk, C., Peterson, P., Landsfeld, M., Pedreros, D., Verdin, J., Shukla, S., Husak, G., Rowland, J., Harrison, L., Hoell, A., & Michaelsen, J. (2015). The climate hazards infrared precipitation with stations—A new environmental record for monitoring extremes. *Scientific Data*, *2*, 150066. https://doi.org/10.1038/sdata.2015.66

Gibbs, I., & Candès, E. (2021). Adaptive conformal inference under distribution shift. *Advances in Neural Information Processing Systems*, *34*, 1660–1672.

Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D., & Moore, R. (2017). Google Earth Engine: Planetary-scale geospatial analysis for everyone. *Remote Sensing of Environment*, *202*, 18–27. https://doi.org/10.1016/j.rse.2017.06.031

Hoo, S. B., Müller, S., Salinas, D., & Hutter, F. (2025). *From tables to time: Extending TabPFN-v2 to time series forecasting*. arXiv. https://arxiv.org/abs/2501.02945

Huffman, G. J., Stocker, E. F., Bolvin, D. T., Nelkin, E. J., & Tan, J. (2023). *GPM IMERG final precipitation L3 half hourly 0.1 degree x 0.1 degree V07*. NASA Goddard Earth Sciences Data and Information Services Center. https://doi.org/10.5067/GPM/IMERG/3B-HH/07

Knapp, K. R., Kruk, M. C., Levinson, D. H., Diamond, H. J., & Neumann, C. J. (2010). The International Best Track Archive for Climate Stewardship (IBTrACS): Unifying tropical cyclone data. *Bulletin of the American Meteorological Society*, *91*(3), 363–376. https://doi.org/10.1175/2009BAMS2755.1

Lehner, B., & Grill, G. (2013). Global river hydrography and network routing: Baseline data and new approaches to study the world's large river systems. *Hydrological Processes*, *27*(15), 2171–2186. https://doi.org/10.1002/hyp.9740

Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, *30*, 4765–4774.

Matsushita, N., Ng, C. F. S., Kim, Y., Suzuki, M., Saito, N., Ariyoshi, K., Salva, E. P., Dimaano, E. M., Villarama, J. B., Go, W. S., & Hashizume, M. (2018). The non-linear and lagged short-term relationship between rainfall and leptospirosis and the intermediate role of floods in the Philippines. *PLoS Neglected Tropical Diseases*, *12*(4), e0006331. https://doi.org/10.1371/journal.pntd.0006331

OpenStreetMap contributors. (2026). *OpenStreetMap data extract: Philippines*. Geofabrik. https://download.geofabrik.de/asia/philippines.html

Page, L., Brin, S., Motwani, R., & Winograd, T. (1999). *The PageRank citation ranking: Bringing order to the web* (Technical Report No. 1999-66). Stanford InfoLab.

Panagopoulos, G., Nikolentzos, G., & Vazirgiannis, M. (2021). Transfer graph neural networks for pandemic forecasting. *Proceedings of the AAAI Conference on Artificial Intelligence*, *35*(6), 4838–4845. https://doi.org/10.1609/aaai.v35i6.16616

Petricola, S., Reinmuth, M., Lautenbach, S., Hatfield, C., & Zipf, A. (2022). Assessing road criticality and loss of healthcare accessibility during floods: The case of Cyclone Idai, Mozambique 2019. *International Journal of Health Geographics*, *21*, 14. https://doi.org/10.1186/s12942-022-00315-2

Philippine Statistics Authority. (2021). *Philippines: 2020 census of population and housing, population by municipality*. Humanitarian Data Exchange. https://data.humdata.org/dataset/c1edf0c5-58c8-4d39-b566-937b1ad7907f

Philippine Statistics Authority & National Mapping and Resource Information Authority. (2025). *Philippines: Subnational administrative boundaries (COD-AB)*. Humanitarian Data Exchange. https://data.humdata.org/dataset/caf116df-f984-4deb-85ca-41b349d3f313

Runge, J. (2020). Discovering contemporaneous and lagged causal relations in autocorrelated nonlinear time series datasets. *Proceedings of the 36th Conference on Uncertainty in Artificial Intelligence*, *124*, 1388–1397. https://arxiv.org/abs/2003.03685

Runge, J., Nowack, P., Kretschmer, M., Flaxman, S., & Sejdinovic, D. (2019). Detecting and quantifying causal associations in large nonlinear time series datasets. *Science Advances*, *5*(11), eaau4996. https://doi.org/10.1126/sciadv.aau4996

Tariverdi, M., Nunez-del-Prado, M., Leonova, N., & Rentschler, J. (2023). Measuring accessibility to public services and infrastructure criticality for disasters risk management. *Scientific Reports*, *13*, 1569. https://doi.org/10.1038/s41598-023-28460-z

Tatem, A. J. (2017). WorldPop, open data for spatial demography. *Scientific Data*, *4*, 170004. https://doi.org/10.1038/sdata.2017.4

Tellman, B., Sullivan, J. A., Kuhn, C., Kettner, A. J., Doyle, C. S., Brakenridge, G. R., Erickson, T. A., & Slayback, D. A. (2021). Satellite imaging reveals increased proportion of population exposed to floods. *Nature*, *596*, 80–86. https://doi.org/10.1038/s41586-021-03695-w

Tiecke, T. G., Liu, X., Zhang, A., Gros, A., Li, N., Yetman, G., Kilic, T., Murray, S., Blankespoor, B., Prydz, E. B., & Dang, H.-A. H. (2017). *Mapping the world population one building at a time*. arXiv. https://arxiv.org/abs/1712.05839

UN-SPIDER. (n.d.). *Recommended practice: Flood mapping and damage assessment using Sentinel-1 SAR data in Google Earth Engine*. United Nations Office for Outer Space Affairs. https://www.un-spider.org/advisory-support/recommended-practices/recommended-practice-google-earth-engine-flood-mapping

van Panhuis, W. G., Choisy, M., Xiong, X., Chok, N. S., Akarasewi, P., Iamsirithaworn, S., Lam, S. K., Chong, C. K., Lam, F. C., Phommasak, B., Vongphrachanh, P., Bouaphanh, K., Rekol, H., Hien, N. T., Thai, P. Q., Duong, T. N., Chuang, J.-H., Liu, Y.-L., Ng, L.-C., et al. (2015). Region-wide synchrony

and traveling waves of dengue across eight countries in Southeast Asia. *Proceedings of the National Academy of Sciences*, *112*(42), 13069–13074. https://doi.org/10.1073/pnas.1501375112

Weiss, D. J., Nelson, A., Vargas-Ruiz, C. A., Gligorić, K., Bavadekar, S., Gabrilovich, E., Bertozzi-Villa, A., Rozier, J., Gibson, H. S., Shekel, T., Kamath, C., Lieber, A., Schulman, K., Shao, Y., Qarkaxhija, V., Nandi, A. K., Keddie, S. H., Rumisha, S., Amratia, P., et al. (2020). Global maps of travel time to healthcare facilities. *Nature Medicine*, *26*, 1835–1838. https://doi.org/10.1038/s41591-020-0931-3

Zadrozny, B., & Elkan, C. (2002). Transforming classifier scores into accurate multiclass probability estimates. *Proceedings of the 8th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 694–699. https://doi.org/10.1145/775047.775151

A Acronyms

Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS)
Daily satellite-and-gauge rainfall on a grid of about 5 km; the main rain input.

Common Operational Dataset of Administrative Boundaries (COD-AB)
The current boundary set; its codes are the join key.

Copernicus Emergency Management Service (EMS)
Rapid flood mapping used to check Sentinel-1 for Ompong.

Department of Health (DOH)
Publisher of the weekly dengue series that is the forecast target.

False discovery rate (FDR)
The Benjamini–Hochberg control applied to the lead-lag links.

High Resolution Settlement Layer (HRSL)
Population grids by age group, used for under-5 and 60+ people.

Humanitarian Data Exchange (HDX)
The open data portal hosting the dengue, census, boundary and city series.

Humanitarian OpenStreetMap Team (HOT)
Publisher of the OSM health facility extract used as hospitals.

Incidence rate ratio (IRR)
The multiplicative change in cases per unit of exposure in the leptospirosis models.

Integrated Multi-satellitE Retrievals for the Global Precipitation Measurement mission (IMERG)
Satellite rainfall, the second rain input.

International Best Track Archive for Climate Stewardship (IBTrACS)
Storm tracks and gale radii.

Japan Meteorological Agency (JMA)
Source of the 30-knot radius used when the United States radius is missing.

Malaria Atlas Project (MAP)
Publisher of the friction surface behind the raster access method.

Message-passing neural network with long short-term memory (MPNN-LSTM)
The graph neural network benchmark.

National Capital Region (NCR)
Metro Manila, the access benchmark.

Nationwide Operational Assessment of Hazards (NOAH)
Flood hazard maps per province; the flood proxy.

OpenStreetMap (OSM)
: The road network and part of the hospital list.

Philippine Integrated Disease Surveillance and Response (PIDSR)
: The surveillance system behind the city leptospirosis series.

Philippine Standard Geographic Code (PSGC)
: The national place code; COD-AB codes follow it.

Roll-on/roll-off (RoRo)
: Vehicle ferries; their routes are edges of the access network.

Shapley additive explanations (SHAP)
: Per-forecast feature attributions shown as drivers.

Vertical-horizontal (VH)
: The Sentinel-1 polarisation used for flood detection.

Weighted interval score (WIS)
: The proper score of a quantile forecast; lower is better.